SIMPLE CONNECTIVITY OF THE LIDMAN–PICCIRILLO PIECE: A CERTIFICATE-COMPLETE COMPUTATION

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ABSTRACT. Let V be the symplectic homology $S^2 \times D^2$ constructed by Lidman and Piccirillo (arXiv:2505.14387) from a genus-2 surface bundle over a punctured torus by two Luttinger surgeries. We compute the fundamental groups of V and of all its admissible surgery variants from a fully explicit based model of the bundle: a rotation-symmetric octagon model of the fiber in which every monodromy lift is genuinely based and every meridian, lasso and Lagrangian-framing word is derived mechanically from finite combinatorial certificates. The resulting presentations are proved complete via a graph-of-groups decomposition and decided by coset enumeration and Knuth–Bendix completion: for all nine admissible surgery cells, including the Lidman–Piccirillo surgery itself, the fundamental group is trivial. The two cells that resist every coset enumeration — past 10^8 cosets — are decided by confluent rewriting systems whose rules reduce every generator to the identity; the same certificates re-derive the entire grid and pass an independent second-diagram cross-check. Combined with the reduction theorem of the companion paper, this yields — from the Lidman–Piccirillo piece and from every admissible variant — an exotic $S^2 \times S^2$, the first pair of homeomorphic closed 4-manifolds distinguished by unconstrained knot slicing, and a simply connected exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. Because based-loop computations in this setting are delicate — the companion paper shows that triviality is the generic outcome of incorrectly derived correction words — every input word carries a machine-checkable certificate, the pipeline is calibrated on a known-answer double-Luttinger configuration on T^4 (Baldridge–Kirk), where it reproduces the published non-abelian fundamental groups in every convention and detects single-whisker errors, and all verdicts reproduce in minutes from the ancillary files.

1. INTRODUCTION

Lidman and Piccirillo [LP25] constructed spin rational homology 4-spheres B and W with $H_1 = \mathbb{Z}/2$ and isomorphic integer cohomology rings such that the figure-eight knot is slice in B and not in W. Their key piece is a symplectic homology $S^2 \times D^2$, here denoted V, obtained from the genus-2 surface bundle R over a once-punctured torus with monodromies φ (along α) and ab (along β) by $(+1)$-Luttinger surgeries on two Lagrangian tori $T_\alpha = (c, \alpha')$ and $T_\beta = (e, \beta')$; then $W = V/\sigma$ for a free boundary involution σ. In the companion paper [W26] we showed that the natural upgrade of their theorem — a *homeomorphic* pair distinguished by slicing — is gated: for any admissible surgery variant V' (classified there), $W' \cong_{\text{homeo}} B$ if and only if $\pi_1(V' \cup_\sigma V') = 1$, in which case the double is an exotic $S^2 \times S^2$; and if $\pi_1(V') = 1$ one additionally obtains a simply connected exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. The value of $\pi_1(V')$ therefore yields three theorems, two of them long-standing open problems.

The companion paper also quantified why this computation is error-prone. The candidate presentations of $\pi_1(V')$ depend on based-loop data — meridian, lasso and Lagrangian-framing words — that free-homotopy reasoning cannot determine; and across 120 sampled candidate correction words, 58 presented the trivial group, 55 a group not visibly trivial, and the

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remaining 7 were detectably wrong ($H_1 \neq 0$): *both nondegenerate outcomes are generic*, so a trivial verdict obtained from guessed words is worthless. The same diagnosis applies to the one prior claim of an exotic $S^2 \times S^2$, by Akhmedov and Park [AP10]: their group theory is machine-verifiably correct given the presentation printed in their Lemma 8, and what has kept that preprint unaccepted for sixteen years is precisely the geometric derivation of its based words, which cannot be checked from a figure.

This paper executes the computation with the level of explicitness that the above diagnosis demands. Everything is derived from one fully explicit based model, mechanically, along the following lines (the process was pre-registered before any derivation: method, validation battery and outcome criteria were fixed in advance):

(1) **Certificates, not figures.** Every curve and every path is specified by a finite combinatorial certificate (a crossing sequence in a fixed octagon model with a fixed vertex-link structure), and its based word is computed by a small developing-map engine, not read off a picture. The engine itself is validated against known answers (Section 3).
(2) **Convention choices are swept or derived.** All meridian signs, correction placements, direction orientations and fiber-mixing exponents are swept ($\approx$ 1,400 presentations in total); the genuinely non-sweepable convention choices (membrane routing; the choice of lasso arc; the sign-coupling of the pushoff-basing correction, Section 4.5) are resolved by derivation and cross-checked by a second, independently derived diagram (Section 7).
(3) **Completeness, not plausibility.** A graph-of-groups decomposition of the surgered manifold identifies the complete relation set; it exceeds the naive one by exactly one relation, which we derive and add, and the corner-region interaction of the two cut systems is shown to contribute nothing further (Section 6).
(4) **Checkability.** Section 9 is a verification guide: it lists the critical steps of the derivation chain, in decreasing order of delicacy, with the certificate to check for each.

Theorem 1.1 (Main result)**.** *Let $V'_{m,n}$ denote the admissible surgery variants of V (fiber-mixing exponents $m, n \in \{-1, 0, 1\}$; the Lidman–Piccirillo piece is $V = V'_{0,0}$). For all nine cells,*

$$\pi_1(V'_{m,n}) = 1.$$

Six cells are decided by coset enumeration; the $n = +1$ column, whose enumerations exceed 10^8 cosets without terminating, is decided by Knuth–Bendix completion: the confluent rewriting system of each presented group reduces every generator to the identity. The completion certificates in fact re-derive all nine cells, in every sign convention, in both diagrams (Section 7).

The proof occupies Sections 2–7: the complete presentations are assembled in Sections 2–6 and decided by coset enumeration and Knuth–Bendix completion in Section 7; every presented group has $H_1 = 0$, as the homology analysis of the companion paper requires, and the verdicts hold in every sign, placement and diagram convention swept there.

Corollary 1.2. *$Z = V \cup_\sigma V$ is an exotic $S^2 \times S^2$; the Lidman–Piccirillo pair (B, W) is a homeomorphic-but-not-diffeomorphic pair of closed 4-manifolds distinguished by the sliceness of the figure-eight knot; and the regluing of* [LP25, Theorem 2] *applied to Z yields a simply connected exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.*

This is immediate from Theorem 1.1 and the reduction theorem of the companion paper [W26, Theorem 1.2].

Remark 1.3 (Verification standard)**.** Because the sixteen-year history of this problem turns on exactly the kind of based-loop data computed here, the derivation was carried out under a pre-registered protocol: the method, validation battery and outcome criteria were fixed before any derivation, every convention that could not be derived was swept, and every input word is backed by a finite certificate checkable independently of the text. One input word required repair after the first version of this derivation: a known-answer calibration of the full pipeline on a Baldridge–Kirk T^4 configuration (Section 4.6) exposed a missed whisker-sweep crossing in $\mathrm{dir}^{\mathrm{base}}_{T_\beta}$; the repaired word is derived with the same certificate standard (Section 4.5), and every verdict was re-established with it. Section 9 indexes the critical steps for a reader who wishes to verify (or refute) the chain; all verdicts reproduce from the ancillary files in minutes.

Ancillary files. The developing engine (`develop.py`) with its run record, the decision harnesses (`decide.g`, `decide2.g`, `vdiag2.g`, `placement_check.g`, `vr_check.g`), the Knuth–Bendix certifications of the full grid in both diagrams (`kb_certify.g`, `kb_diag2_full.g`; require the `kbmag` package [KBMAG]) with their run records, the T^4 known-answer calibration with its pre-registered design and run record (`decide_t4.g`; Section 4.6), the independent MAF cross-check (`maf_export.g`, `maf_certify.sh` [MAF]) with its run record and the staged corrected-word exports (`maf_export2.g`), the deep-enumeration record (`tc_deep.g`), the finite-quotient harnesses (`phase2_*`, `phase3_*`) and all run logs. Every number in this paper reproduces from them; total runtime is minutes on a laptop (GAP 4.16 [GAP], Python 3), except the optional quotient sweep (≈56 CPU-hours). The same files are also available at `https://github.com/bwuebben/exotic-s2xs2`.

2. The based model

2.1. **The symmetric octagon.** The fiber F is the regular octagon with edge word $x\,y\,x^{-1}y^{-1}\,r\,s\,r^{-1}s^{-1}$ (edges $E_1,\dots,E_8$, vertices $V_1,\dots,V_8$, all identified to the single point p), so $\pi_1(F,p)=\langle x,y,r,s \mid [x,y][r,s]\rangle$ with the edge loops as generators. Let ρ be the rotation by π. It maps $E_i\to E_{i+4}$, respects the identifications, and induces an involution φ_0 of F with: φ_0 acts on $\pi_1(F,p)$ by the *exact* swap $x\leftrightarrow r$, $y\leftrightarrow s$ — with no *whisker*, our term throughout for a basing arc back to p and the conjugation it imposes on based words; and $\mathrm{Fix}(\varphi_0)=\{p,O\}$ (O the center) — matching the two fixed points of the Lidman–Piccirillo involution φ and giving their two sections Γ,Γ' through p and O.

Since φ_0 fixes p and the twist diffeomorphism ψ_0 below is supported away from p, *every monodromy lift in this paper is a genuinely based automorphism*: the presentation of π_1 of the bundle carries no whisker debt. This eliminates one entire class of based-loop ambiguity before the computation starts.

2.2. **Curves and lifts.** The twist curves are a, b (push-offs of the x- and y-circles, crossing the y- resp. x-circle once near p) and their ρ-images $e:=\varphi_0(a)$, $d:=\varphi_0(b)$, so Lidman–Piccirillo's symmetry ($\varphi\colon a\leftrightarrow e$, $b\leftrightarrow d$) is built into the model. The twist directions are normalized so that the based actions are $T_a\colon(x,y)\mapsto(x,yx)$, $T_b\colon(x,y)\mapsto(xy^{-1},y)$ — a twist direction is a binary choice per curve, some choice realizes these standard once-punctured-torus formulas, and the residual chirality ambiguity is absorbed by the handle swap [W26, Remark 2.3]; the composite $h:=T_a\circ T_b\colon x\mapsto y^{-1},\ y\mapsto yx$ is machine-certified to fix the boundary word $[x, y]$ exactly, to satisfy $h^6 = \mathrm{conj}_{[x,y]^{-1}}$ (the boundary Dehn twist) and $h^3 = -\mathrm{id}$ on H_1: the trefoil monodromy. Set $\tilde{\psi} := h * \mathrm{id}$ (well defined on the closed-fiber group because h fixes $[x, y]$ exactly) and $\tilde{\varphi} :=$ the swap; the identity $\tilde{\psi}\tilde{\varphi}\tilde{\psi}^{-1}\tilde{\varphi}^{-1} = h * h^{-1}$ is machine-certified and realizes Lidman–Piccirillo's factorization $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$. The diffeomorphism $\psi_0 := T_aT_b$ fixes p, O and the entire right handle pointwise; in particular $\psi_0(e) = e$ pointwise.

The invariant curves are forced by homology and ρ-symmetry to be two-arc curves: $c = \gamma \cup \rho(\gamma)$ with γ an embedded arc from the y-edge pair to the s-edge pair, and z likewise on the x/r-pairs. Their based words are *derived*, not assumed, in Section 4.

2.3. **The cut-square model of the bundle.** The base T_0 (once-punctured torus) is the square $[0, 1]^2$ minus a puncture disk at $(3/4, 3/4)$, with $(\xi, (t, 1)) \sim (\psi_0\xi, (t, 0))$ and $(\xi, (1, u)) \sim (\varphi_0\xi, (0, u))$: crossing the α-cut upward applies ψ_0; crossing the β-cut rightward applies φ_0. Basepoint (p, q) with $q = (1/4, 1/4)$; based base loops $\bar{\alpha}$ (horizontal through q; holonomy $\tilde{\varphi}$) and $\bar{\beta}$ (vertical; holonomy $\tilde{\psi}$), realizing the minimal crossing pattern with the embedded curves: $\bar{\alpha}$ meets the embedded β once and the embedded α not at all, and symmetrically for $\bar{\beta}$ — minimal because $|\alpha \cdot \beta| = 1$ forces one crossing, while T_0 cut along α still contains a boundary-parallel copy of α basable at q with trivial whisker. The surgery tori sit on the cuts: $T_\alpha = c \times \{\alpha\text{-cut}\}$ (closing because $\varphi_0(c) = c$; the restriction $\varphi_0|_c$ is a free half-rotation — a framing fact used in Section 4.4) and $T_\beta = e \times \{\beta\text{-cut}\}$ (closing with the product framing because $\psi_0(e) = e$ pointwise).

Each monodromy relation of the bundle is carried by a *membrane*. Concretely, for a fiber generator g transported around $\bar{\beta}$, the membrane is the map of a square

$$\mathfrak{M}_g(u, v) = \big(\text{transport of } g(u) \text{ along } \bar{\beta} \text{ to time } v\big) :$$

its bottom edge is g, its top edge is the transported loop $\tilde{\psi}(g)$, and both vertical edges are $\bar{\beta}$ itself (the track of the basepoint, since the monodromies fix p). In the unsurgered bundle R this square *is* the homotopy behind the monodromy relation $BgB^{-1} = \tilde{\psi}(g)$. In the surgered manifold the membrane may meet the surgery tori, and each transverse intersection point punctures it: the relation then holds only up to one conjugated-meridian factor per puncture, $BgB^{-1} = (\ell\,\mu^{\pm 1}\ell^{-1}) \cdots \tilde{\psi}(g)$, where ℓ is the in-membrane path from the base corner to the puncture (Section 4.3 turns each such factor into an explicit word). Because each surgery torus is a product (fiber curve) $\times$ (base cut circle), the punctures are exactly the points (base crossing of the loop with the cut) $\times$ (fiber crossing of g with the torus' fiber curve) — a finite count, read off the position tables of Section 4.2.

Remark 2.1 (A worked membrane)**.** Take $g = y$ over $\bar{\beta}$, the simplest corrected relation. The base track of the membrane is the vertical circle through $q = (1/4, 1/4)$: traveling upward from q, it crosses the α-cut exactly once, at the wrap $(1/4,\ 1{\equiv}0)$, and never meets the β-cut (its column stays at $u_1 = 1/4$). So this membrane can meet only $T_\alpha = c \times \{\alpha\text{-cut}\}$, and it does so exactly where its fiber point lies on c: the loop y crosses c once, at c_y (position table), so the membrane meets T_α in the single transverse point $(c_y,\ (1/4, 1{\equiv}0))$. Puncturing there, the clean relation $ByB^{-1} = \tilde{\psi}(y) = yx$ acquires exactly one conjugated-meridian factor, whose lasso is the in-membrane path to the puncture: the partial loop y_1 (from p to c_y) followed by the vertical transport up to the cut. Defining the based meridian M *along that very whisker* ($M := y_1 \cdot (\text{meridian circle at } c_y) \cdot y_1^{-1}$, Section 4.3) absorbs the lasso entirely, and the corrected relation is $ByB^{-1} = M^{\pm 1}(yx)$. Every other membrane is this computation with different position data. Over $\bar{\beta}$: the loop s crosses c once at $P_{s'}$, giving

$BsB^{-1} = (\delta M^{\pm 1}\delta^{-1})s$ with the explicit lasso δ (the meridian is based at c_y, so the basing must slide along c from $P_{s'}$ — that slide is δ); the loops x and r miss c entirely, so their relations stay clean. Over $\bar{\alpha}$: the base track (the horizontal circle through q) crosses the β-cut once and never meets the α-cut, so only $T_\beta = e \times \{\beta\text{-cut}\}$ is in play; the loop s crosses e once at s_e, giving $AsA^{-1} = N^{\pm 1}y$ (with N based along the s_2-whisker, again absorbing its own lasso), while x, y, r miss e, so their relations stay clean. That is the full count: three corrected relations, five clean ones, each correction at a single crossing point. The same point-versus-circle discipline governs the *pushoff* basings — there the object transported around $\bar{\beta}$ is a whisker rather than a generator, and its membrane is subject to the same crossing count (Section 4.5).

3. The developing engine and its validation

A based loop transverse to the edge circles is specified by a *certificate*: the departure corner wedge V_i, the ordered list of edges crossed, and the arrival wedge V_j. In the universal cover tiled by octagons, the tile with corner V_i at the base lift is $w_i^{-1}D_0$, where w_i is the length-$(i-1)$ prefix of the relator; crossing out of a tile through its edge E_i composes the deck element with $g_{\mathrm{out}}(i) = w_i\, w^{-1}_{\mathrm{partner}(i)+1}$ ($E_1{\sim}E_3$, $E_2{\sim}E_4$, $E_5{\sim}E_7$, $E_6{\sim}E_8$, reversed). The based word of a certificate is $w_i^{-1}\big(\prod g_{\mathrm{out}}\big)w_j$, reduced by Dehn's algorithm for the genus-2 relator (surface groups have Dehn presentations, so the reduction decides the word problem). The engine (`develop.py`) implements exactly this and nothing else.

Validations (all machine-checked; output in the ancillary run record).

- V1. Every edge-parallel certificate returns its generator $(x, y, x^{-1}, r, s, s^{-1})$.
- V2. The vertex-link loop — crossing all eight edges in the order $E_1E_4E_3E_2E_5E_8E_7E_6$ dictated by the link walk $V_1V_4V_3V_2V_5V_8V_7V_6$ — develops to 1: a single global identity constraining all eight decks and the link structure simultaneously.
- V3. The push-off certificates return $a \simeq x^{-1}$, $b \simeq y$, $e \simeq r^{-1}$, $d \simeq s$, exactly swap-equivariantly.
- V4. The two lasso routes of Section 4.3 differ by a whiskered copy of the closed curve c — the slide-around-the-curve identity — exactly.

4. The derived words

4.1. **The invariant curves.** The curve c (certificate: cross out E_8, then out E_4) has based word $(rx)^{-1}$ at the V_2-whisker; the engine's whisker table (c developed at the V_1, V_2, V_3, V_5 whiskers; run record) realizes the xr/rx ordering ambiguity as pure whisker choice. The ρ-symmetric z (certificate $[E_7, E_3]$) has free class sy. Both reproduce the free-homotopy dictionary of [W26] and sharpen it: the orderings that the companion paper's experiment E1 had to sweep are now derived.

4.2. **Position tables.** Disjointness pins the normal positions: on the s-edge, the c-crossing precedes the e-crossing (else $c \cap e \neq \varnothing$); on the y-edge only the c-crossing is relevant. Realizability of the full configuration $\{a, b, c, d, e, z\}$ with Lidman–Piccirillo's intersection pattern is checked against the certificates (c meets d's strip on leaving E_6 and b's strip on leaving E_2; z meets e's strip once at its E_7-crossing; all curves avoid p and O).

4.3. **Meridians, lassos, corrections.** Let y_1 and s_1 be the initial segments of the based loops y and s up to their c-crossings, and s_2 the initial segment of s up to its e-crossing (on the s-edge the c-crossing comes first, by the position table). Base the meridians at the surgery-relevant crossings: $M := y_1\cdot(\text{meridian circle of } T_\alpha)\cdot y_1^{-1}$, $N := s_2\cdot(\text{meridian circle of } T_\beta)\cdot s_2^{-1}$. Then the membrane analysis gives the three corrected relations in the form

$$ByB^{-1} = M^{\pm}\,(yx), \qquad BsB^{-1} = \left(\delta M^{\varepsilon}\delta^{-1}\right) s, \qquad AsA^{-1} = N^{\pm}\, y,$$

i.e. two of the three corrections are *exactly* meridian powers (the basing absorbs the lasso), and the third has lasso $\delta = s_1 \cdot (\text{arc of } c)\cdot y_1^{-1}$, computed by the engine: $\delta = r^{-1}$ (via one arc of c) or $\delta' = x$ (via the other), with the validation V4 identity $\delta'\delta^{-1} = xr \sim c$. All signs are swept ($\varepsilon \in \{\pm 1\}$ denotes the swept sign of the Bs-correction, which reappears in Section 4.5); the left placement of the corrections is tied to the stated membrane orientation and is probed by a full placement sweep (Section 7). The crossing $z_0 = (P_{s'}, \text{the wrap of } \bar\beta)$ that produces the Bs-correction corrects one more word: the base-direction pushoff of T_β (Section 4.5).

4.4. **The direction (Lagrangian-framing) words.** For a loop on a surgery torus traveling once in the base direction and crossing one cut with holonomy Θ at fiber position η, with σ a fiber path from p to η: the based pushoff is $(\text{cut generator})\cdot\tilde\Theta(\sigma)\cdot(\text{fiber drift})\cdot\sigma^{-1}$. This yields:

$\mathrm{dir}^{\mathrm{base}}_{T_\beta} = \left(\delta M^{-\varepsilon}\delta^{-1}\right) B$	(ψ_0 fixes the right handle pointwise: zero drift; the meridian conjugate is the pushoff-basing correction, Section 4.5),
$\mathrm{dir}^{\mathrm{fib}}_{T_\beta} = s\, r^{-1} s^{-1}$	(e at the s_2-basing; $= s\,(e@V_7)\,s^{-1}$ ✓, $e@V_7$ the engine's development of e departing from wedge V_7),
$\mathrm{dir}^{\mathrm{base}}_{T_\alpha} = A\, r^{-1}$	(the half-rotation drift is *exactly* the lasso: $\varphi_0(c_y) = c_s$, $\tilde\varphi(y_1) = s_1$),
$\mathrm{dir}^{\mathrm{fib}}_{T_\alpha} = (rx)^{-1}$	(c at the y_1-basing).

(Here c_y and c_s denote c's crossing points on the y- and s-edges; $\varphi_0(c_y) = c_s$ says the half-rotation carries one to the other, which with $\tilde\varphi(y_1) = s_1$ is exactly why the drift equals the lasso.) Because $\varphi_0|_c$ is a free half-rotation, the base direction on T_α cannot close without drifting along half of c: write λ for the resulting closed drift curve — the base-direction pushoff itself. The drift identity $\lambda^2 = x^{-1}A^2r^{-1}$, i.e. $2\lambda = 2\alpha' + [c]$ in homology, holds exactly and is the sharpest internal check on this, the single most delicate input word. The (m,n)-family directions are $\mathrm{dir}^{\mathrm{base}}\cdot(\mathrm{dir}^{\mathrm{fib}})^{m \text{ or } n}$; the two lasso-arc conventions differ by exactly one unit of n, hence are absorbed by the sweep.

4.5. **The pushoff-basing correction.** The first line of the display corrects the one input word of this computation that an earlier version of the derivation got wrong. The pushoff formula above is a bundle identity; realizing it in the *complement* of the surgery tori transports the whisker $\sigma = s_2$ once around $\bar\beta$, sweeping the membrane

$$\mathfrak{A}(u,v) = \big(\text{transport of } s_2(u) \text{ along } \bar\beta \text{ to time } v\big),$$

a square whose sides are s_2, the pushoff's base circle at fiber s_e, $\tilde\psi(s_2) = s_2$ (pointwise — ψ_0 fixes the right handle), and $\bar\beta$.

The membrane misses T_β: its base column never meets the β-cut, and its interior fiber points $s_2(u)$, $u < u_e$, miss e (position table: one e-crossing on the s-edge, at the endpoint of s_2). It crosses T_α at exactly one interior point $z_0 = (P_{s'}, \text{the wrap of } \bar\beta)$: the base circle crosses the α-cut once, and s_2 crosses c exactly once, at $P_{s'}$ — the position table's own ordering (the c-crossing precedes the e-crossing on the s-edge) places that crossing *inside* s_2. The local intersection number is ± 1, so no interior homotopy avoids it. The justification given in the display's first version ("the position homotopy avoids T_α since $c \cap e = \varnothing$") is true of — and only of — the constant-fiber column slide of the pushoff circle, whose fiber point $s_e \in e$ indeed never lies on c; it says nothing about the whisker sweep. Testing a circle's transport by the crossings of a single point is precisely the error class that the known-answer calibration of Section 4.6 detects.

Puncturing the (disk) membrane at z_0 and reducing by the *same* steps that produced the Bs-correction — the crossing point, the in-membrane lasso and the arc route are literally those of Section 4.3, because $\mathfrak{A}$ is the restriction of the Bs-membrane $s \times \bar{\beta}$ to $u \leq u_e$ — gives, with ε the sign of the Bs-correction in the same convention,

$$\mathrm{dir}^{\mathrm{base}}_{T_\beta} = \left(\delta M^{-\varepsilon} \delta^{-1}\right) B, \qquad \delta = r^{-1} :$$

the B-conjugation in the membrane identity cancels exactly (so the insertion is on the left, with the plain lasso δ), and the meridian sign is *anti-coupled* to the Bs-correction sign. Both are relative statements, hence covariant under the global orientation flip that the sign sweep blankets; no new swept convention is introduced. The correction abelianizes to zero, so the homology bookkeeping of [W26, Lemma 3.1] is unchanged; and the arc choice is immaterial in the group, since $\delta' M \delta'^{-1}$ and $\delta M \delta^{-1}$ differ by conjugating M by a whiskered copy of c (the V4 identity), a pushoff class on the boundary 3-torus $\partial\nu(T_\alpha)$, which commutes with the meridian there.

The corrected, sign-coupled word changes no verdict: the full battery — the grid sweep, the placement probe, the second diagram, and the Knuth–Bendix certification — was rerun with it, and every admissible cell remains trivial in every convention (Section 7). Moreover, before the exact derivation was fixed, all eight candidate resolutions of the correction (both arcs, both signs, both placements) were swept at the surgery cell $(0, 0)$ — all 256 runs certified trivial (enumeration plus rewriting certificates) — and both arc candidates were swept across the full completed-presentation grid: 576/576 certified trivial. The theorem never hinged on the resolution; only the derivation's honesty did.

4.6. **Known-answer calibration: the Baldridge–Kirk T^4 configuration.** The pipeline that produces and decides these presentations — the cut-model conventions, the membrane crossing counts, the meridian basings and lassos, the filling format, the completeness recipe — was calibrated on a configuration whose answer is known independently: the disjoint Lagrangian tori $T_1 = X \times A_1$, $T_2 = Y \times A_2$ in T^4 of Baldridge and Kirk [BK09], whose complement presentation they computed with explicit care about basings, and for which double ± 1 Luttinger surgery gives $\pi_1 = (\mathbb{Z}^2 \rtimes_A \mathbb{Z}) \times \mathbb{Z}$ with $|\operatorname{tr} A| \in \{1, 3\}$ according to the relative signs — non-abelian in every convention. Deriving the input words for this configuration by the same membrane method and sweeping all 64 sign/placement conventions: with the correctly derived lasso data the harness reproduces the ground truth in all 64 cases, with the exact $\mathrm{tr} = 3/\mathrm{tr} = 1$ fingerprint split; with every wrong lasso pair (including the chirality flip) it fails all 64, half the failures abelian — the signature of a missed whisker crossing. A single surgery reproduces $H_3(\mathbb{Z}) \times \mathbb{Z}$ in all conventions, and Baldridge–Kirk's published words, run through the same decision battery, match ours cell by cell. The instrument detects single-whisker errors — and it is what found the correction of Section 4.5: the first, hand-derived T^4 words failed exactly as the test's pre-registered criteria describe, the diff against the published presentation localized the error to one missed circle-slide crossing, and re-auditing this paper's derivation for the same error class found it in — and only in — $\mathrm{dir}^{\mathrm{base}}_{T_\beta}$ (the symmetric candidates come out clean: $y_1 \cap e = \varnothing$ protects $\mathrm{dir}^{\mathrm{base}}_{T_\alpha}$, and $\mathrm{dir}^{\mathrm{fib}}_{T_\beta}$'s whisker tail is crossing-free). The harness, its pre-registered design and the run record are in the ancillary files. One scope note: T^4 has trivial monodromy, so the calibration does not exercise the monodromy-drift machinery behind $\mathrm{dir}^{\mathrm{base}}_{T_\alpha} = Ar^{-1}$; that word's audit is Lemma 4.1 and item 1 of Section 9.

4.7. **The framing lemma.** Luttinger surgery is performed with respect to the *Lagrangian framing*: in a Weinstein chart — a symplectomorphism of $\nu(T)$ with a neighborhood of the zero section of T^*T^2, canonical coordinates $(\Theta_1, \Theta_2; P_1, P_2)$, $\omega = dP_1 \wedge d\Theta_1 + dP_2 \wedge d\Theta_2$, $T = \{P = 0\}$ — the framing pushoff of a curve on T is its lift at *constant momentum*. Two framings of T differ by adding meridian multiples to the pushoff classes, and a meridian error in a direction word changes the surgery coefficient; this is why the framing cannot be swept and was identified as the principal risk. The following lemma closes it.

Lemma 4.1 (Framing)**.** *Fix the Thurston-type symplectic form on R built from a fiber area form invariant under both monodromies, chosen in the normalized local models below; this is a form of the kind used in* [LP25, Section 1]*. Then for both T_β and T_α the fibered pushoffs used in Section 4.4 — the in-fiber parallel copy of the fiber curve, and the base-direction shift of the closed base-direction curve — are constant-momentum lifts in an explicit Weinstein chart. In particular their classes on $\partial\nu(T)$ have zero meridian component: the fibered framing* is *the Lagrangian framing.*

Proof. *T_β.* The twist supports of ψ_0 lie in the left handle, so $\psi_0 = \mathrm{id}$ on a neighborhood of e; hence the bundle is a genuine product over a base annulus near the β-cut. Normalize (Moser, on an annulus) the invariant fiber area form to $dt \wedge d\theta_1$ on $A_e \cong S^1 \times (-1, 1)$ and write the base annulus as (θ_2, u). The Thurston-type form on this chart is $\omega = dt \wedge d\theta_1 + K\, du \wedge d\theta_2$, which is canonical for $(\theta_1, \theta_2;\, t,\, Ku)$, with $T_\beta = \{t = u = 0\}$ the zero section. The fiber pushoff $\{t = \varepsilon, u = 0\}$ and base pushoff $\{t = 0, u = \varepsilon\}$ sit at constant momentum. *T_α.* Here the holonomy around the α-cut is φ_0 with $\varphi_0|_{A_c}$ an area-preserving free half-rotation; normalize it (equivariant Moser) to $(\theta_1, t) \mapsto (\theta_1 + \pi, t)$ with fiber form $dt \wedge d\theta_1$. The ν-region is the quotient of $A_c \times [0, 2\pi] \times (-1, 1)$ by $(\theta_1, t, 2\pi, u) \sim (\theta_1 + \pi, t, 0, u)$, with $\omega = dt \wedge d\theta_1 + K\, du \wedge d\theta_2$ descending to it. The change of coordinates

$$\Theta_1 = \theta_1 - \tfrac{\theta_2}{2}, \qquad \Theta_2 = \theta_2, \qquad P_1 = t, \qquad P_2 = \tfrac{t}{2} + Ku$$

is well defined on the quotient (the deck identification sends $\Theta_1 \mapsto \Theta_1 + 2\pi$) and gives $\omega = dP_1 \wedge d\Theta_1 + dP_2 \wedge d\Theta_2$ with $T_\alpha = \{P = 0\}$: an explicit Weinstein chart on the mapping torus of the half-rotation. Our fiber pushoff $\{t = \varepsilon, u = 0\}$ has $P = (\varepsilon, \varepsilon/2)$; our base pushoff $\{t = 0, u = \varepsilon\}$ — whose base-direction closed curve is precisely the drift curve λ ($\Theta_1 = \mathrm{const}$, i.e. once around the base while drifting half-way along c) — has $P = (0, K\varepsilon)$. Both are constant-momentum lifts. The Lagrangian framing is independent of the choice of Weinstein chart [ADK03], and the based words of Section 4.4 are the engine's developments of exactly these pushoff curves. $\square$

Remark 4.2. The chart re-derives the drift from the symplectic side: in the coordinates above, the closed curves on T_α are generated by the Θ_1-circle ($= c$) and the Θ_2-circle ($= \lambda$); a "pure base" curve does not exist on this torus, so the half-drift in $\mathrm{dir}^{\mathrm{base}}_{T_\alpha} = Ar^{-1}$ is forced twice over — topologically (Section 4.4) and symplectically. The lemma is stated for the exhibited normalized form; [LP25]'s construction permits this choice, all downstream arguments hold for it, and the surgery coefficient's sign convention remains inside the harness sweep.

5. The presentation

Generators x, y, r, s (fiber), A, B (base), M, N (meridians). Relations:

(i) $[x, y][r, s]$;
(ii) clean $\tilde{\varphi}$-monodromy: $AxA^{-1} = r$, $AyA^{-1} = s$, $ArA^{-1} = x$;
(iii) clean $\tilde{\psi}$-monodromy: $BxB^{-1} = y^{-1}$, $BrB^{-1} = r$;
(iv) corrected monodromy, as in Section 4.3 (signs swept);
(v) fillings: $M \cdot (Ar^{-1} \cdot ((rx)^{-1})^n)^{\pm 1}$, $N \cdot \big((\delta M^{-\varepsilon}\delta^{-1})B \cdot (sr^{-1}s^{-1})^m\big)^{\pm 1}$ ($\varepsilon =$ the sign of the Bs-correction in (iv), Section 4.5);
(vi) the completion relation R_3 of Section 6: $B\,(s^{-1}r^{-1}yx)\,B^{-1} = r^{-1}s^{-1}x$.

Known-answer probes: dropping (v) gives $H_1 = \mathbb{Z}^2$; fiber-only fillings give $H_1 = \mathbb{Z}^2$; one filling gives $H_1 = \mathbb{Z}$; the full presentation has $H_1 = 0$ for every cell and sign — each exactly as the homology analysis of [W26, Lemma 3.1] requires.

6. Completeness

The presentation above is derived relation-by-relation, so a priori it only presents a group *surjecting* onto $\pi_1(V')$ (all relations true; completeness unproved), which validates trivial verdicts but not nontrivial ones. We now close this gap.

T_0 retracts onto a neighborhood of the cut wedge, so $\pi_1(C)$ ($C =$ the complement of the surgery tori in R) is the fundamental group of a graph of groups over the cut system: cutting the base along the α-cut gives a pair of pants; the cut collar minus $\nu(T_\alpha)$ is a fibered two-sheet amalgam $\mathrm{MT}(\varphi_0)_- *_{\mathrm{MT}(\varphi_0|_{F \setminus \nu c})} \mathrm{MT}(\varphi_0)_+$; the two collar-to-pants edges form a cycle whose stable letter is B. (The inclusion $\pi_1(F \setminus \nu c) \hookrightarrow \pi_1(F)$ is *not* surjective — F is an HNN extension over it — and with that accounted the amalgam's H_1 matches the Mayer–Vietoris value $\mathbb{Z}^5$, meridian class included; a naive surjectivity assumption fails this cross-check.) Tietze reduction of the graph of groups yields exactly: the relations (i)–(v) *plus* the statement that the clean monodromy relations hold for *every* element of the drilled fiber groups:

$$B\kappa B^{-1} = \tilde{\psi}(\kappa) \ \ \forall \kappa \in \pi_1(F \setminus \nu c), \qquad A\kappa A^{-1} = \tilde{\varphi}(\kappa) \ \ \forall \kappa \in \pi_1(F \setminus \nu e).$$

Two further relation classes reduce to redundancies: the drilled stable-letter relations reduce to the free-group identity $[B, A] = [B, A]$ once one notices that the relevant height-sweep crosses the base puncture; and the outer annulus piece is the mapping torus of $h * h^{-1}$ (not a product — a product reading would force $h * h^{-1}$ to be an inner automorphism, which it is not), whose holonomy relation is a consequence of (ii)–(iii).

It remains to compute free bases of the drilled fiber groups. Cutting the octagon along the arcs of e (one arc, two regions; quotient complex $V{=}3, E{=}7, F{=}2$, $\chi = -2$ ✓) and contracting a spanning tree:

$$\pi_1(F \setminus \nu e,\, p) \text{ is free on } \{x,\, y,\, r\},$$

so (ii)–(iii) already contain the full clean set for T_β: *no completion needed.* Cutting along the two arcs of c (three regions; $V{=}5, E{=}10, F{=}3$, $\chi = -2$ ✓):

$$\pi_1(F \setminus \nu c,\, p) \text{ is free on } \{x,\, r,\, \kappa_3\}, \qquad \kappa_3 = s^{-1}r^{-1}yx$$

(κ_3 = the mid-region boundary-collar loop; certificate: depart V_7, no crossings, arrive V_3; $\langle \kappa_3, [c] \rangle = 0$ ✓). The clean relations (iii) cover only $\{x, r\}$: the completion is the single relation

$$R_3 : \qquad B\,\kappa_3\,B^{-1} = \tilde{\psi}(\kappa_3) = r^{-1}s^{-1}x.$$

R_3 is true by derivation, so adding it can only quotient: trivial verdicts are stable under completion by construction.

Lemma 6.1 (Corner closure)**.** *The interaction region of the two cut collars (over the base point $\alpha \cap \beta$) contributes no relations beyond consequences of* (i)–(vi)*. Hence the completed presentation is a complete presentation of $\pi_1(C)$, and, with the fillings, of $\pi_1(V')$.*

Proof. Decompose iteratively rather than simultaneously. First remove T_α alone: the single-torus analysis above is clean for $C_1 = R \setminus \nu(T_\alpha)$ (there is no second torus to puncture its interfaces), and yields the complete presentation $\pi_1(C_1) = \langle x, y, r, s, A, B, M \mid$ fiber relator; A-clean on all of $\pi_1(F)$; B-clean on $\{x, r, \kappa_3\}$; corrected By, $Bs\rangle$. Now remove T_β from C_1, cutting the base along V. The corner is localized in the interfaces of this second decomposition: the V-parallel interface 3-manifolds are F-bundles over circles drilled at the c-annulus over one fiber (where the H-arc crosses), and the slab-level interface is the $(F \setminus A_e)$-bundle over V drilled at the c-annulus over $\alpha \cap \beta$. Each such drilled interface is a *partial mapping torus*: its fundamental group is $\langle \Pi, \tau \mid \tau\kappa\tau^{-1} = \mathrm{hol}(\kappa)\ \forall \kappa \in \Pi_{\mathrm{drilled}} \rangle$, where the drilled subgroup is $\pi_1(F \setminus \nu c)$ or the doubly-drilled $\pi_1(F \setminus \nu(c \cup e))$. Three observations exhaust the Tietze content. (1) For interface fiber elements, the edge-loop relations of the graph of groups reproduce: the clean A-relations on $\pi_1(F \setminus \nu e)$ (complete by the basis $\{x, y, r\}$), and, for the generators crossing the e-slab, the *corrected As* relation — the corner analysis independently re-derives the correction of Section 4.3, a consistency check it could have failed. (2) The partial-HNN internal relations globalize to $B\kappa B^{-1} = \tilde{\psi}(\kappa)$ for $\kappa \in \pi_1(F \setminus \nu(c \cup e)) \subset \pi_1(F \setminus \nu c)$, a consequence of the completed clean set on $\pi_1(F \setminus \nu c)$ (this is where R_3 is used a second time); no doubly-drilled basis is needed. (3) The stable letters τ of the drilled interfaces can be based at the section: p avoids every torus fiber and is fixed by both monodromies, so their two sheet-readings are pure base words (B, and its puncture-adjusted counterpart), computed with zero drift. The constant-p section $T_0 \hookrightarrow C$ splits the projection $\pi_1(C) \to \pi_1(T_0)$, so $F_2\langle A, B\rangle$ *injects* into $\pi_1(C)$; any corner relation between pure base words is therefore an identity of the free group and contributes nothing. The symmetric statements for the other cut order are identical. Hence every corner relation is included or implied. □

7. Results

The sweep decisions are coset enumerations (cap 4×10^5 cosets); "blowup" means the enumeration exceeded the cap: *not visibly trivial*, never "nontrivial". (The blowups are decided below, by rewriting systems.) Sweeps:

sweep	cases	trivial	blowup	other
initial presentation (i)–(v), all (m, n) and signs	288	116	172	0
placement probe at $(0, 0)$ (2^3 placements $\times$ signs)	256	192	64	0
second diagram (independent Bs-derivation; `vdiag2.g`)	576	8	568	0
completed presentation (adds R_3)	288	192	96	0
Knuth–Bendix certificates (completed grid; second diagram $+R_3$, full grid)	864	864	0	0

Every one of the $\approx 1{,}400$ presented groups has $H_1 = 0$. No case anywhere produced a finite nontrivial group — the outcome that a genuine inconsistency between the diagrams would produce. The corrected pushoff-basing word (Section 4.5) makes the presentations markedly harder for coset enumeration than their pre-repair versions — blowups are more frequent, and in the second diagram nearly universal — but the decisions are unchanged. In the completed presentation, six of the nine (m, n) cells are enumeration-trivial in *all* 32 sign conventions — the Lidman–Piccirillo cell $(0, 0)$ among them — and the $n = +1$ column resists enumeration; the rewriting-system certificates below then decide *every* cell of both diagrams, in every convention. The second diagram (a detour representative of $\bar{\beta}$, whose derivation reduces to a single extra meridian factor $N^{\pm}$ in the Bs relation — the second potential crossing vanishes because $\varphi_0^{-1}(s) \cap e = s \cap a = \varnothing$, and the M-lasso is unchanged because the detour-transport loop is trivial by the position table) agrees with diagram 1 wherever enumeration decides anything, and its full grid is certified below.

The enumeration-resistant cells. The cells $(\pm 1, +1)$ resisted every coset enumeration in every version of the derivation — Tietze simplification reduces each presentation to 3 generators and 7 relators, and the enumerations exceed 8×10^6 cosets in the certification sweeps and past 10^8 in a dedicated deep run — and with the corrected pushoff-basing word the whole $n = +1$ column joins them. A certified quotient sweep (harnesses and run records in the ancillary files) first probed the two hardest cells from below, in eight representative sign conventions (the two uniform and the two alternating patterns per cell, in the pre-repair form of the presentations): the groups are perfect, have no proper subgroup of index at most 7 (low-index enumeration), and admit no epimorphism onto any of the 31 nonabelian simple groups of order at most 10^5 — since a nontrivial finite quotient of a perfect group surjects onto a nonabelian simple group of no larger order, *no nontrivial finite quotient of order at most* 10^5 (≈ 56 CPU-hours of GAP computation). All of these are exactly what the trivial group exhibits when probed from below.

Knuth–Bendix completion (kbmag [KBMAG], shortlex, on the simplified presentations) decides every cell in under a second each: the completion reaches confluence, the language of irreducible words has size one, and *every generator explicitly reduces to the identity*. The last certificate proves triviality independently even of the confluence claim: each rewriting rule is a consequence of the relators by construction, so a reduction $g \to^* 1$ is a derivation of $g = 1$. Controls: the same pipeline builds the automatic structure of the genus-2 surface group and reports $\mathsf{Size} = \infty$; it does *not* complete on the (infinite) group presented by the clean relations alone; and a run that was accidentally given a wrong-convention commutator — hence a different group — certified nothing. With the corrected, sign-coupled word, the certificates decide the *entire* completed-presentation grid (288/288 trivial, agreeing with the enumeration on all 192 cases it decided) and the *entire* second diagram (576/576 — all nine cells, all 64 sign conventions each; triviality verdicts need no completeness, so diagram 2's relation set plus the diagram-independent true relation R_3 suffices; without R_3, Knuth–Bendix is inconclusive on diagram 2's hard cells — no confluence, never an adverse verdict). An independently implemented completion engine, MAF [MAF], reproduced the kbmag verdicts on the eight representative presentations of the pre-repair word (word acceptor accepting exactly one word, every generator reducing to the identity; run record in the ancillary files); the corrected-word exports for the same eight cases are staged in the ancillary files for the identical MAF run. Enumeration blowup past 10^8 cosets on a presentation of the trivial group is a classical phenomenon [HEO05]; it is why "blowup" was never read as “nontrivial”, and both directions of that caution are now borne out in this family: guessed words make triviality generic (the companion’s sensitivity experiment), and correctly derived words can make triviality invisible to enumeration — the corrected word is *harder* to enumerate than the wrong one, while presenting the same trivial group.

8. Consequences

Theorem 1.1 and [W26, Theorem 1.2] together give Corollary 1.2, and give it for *every* admissible cell: each double Z' is an exotic $S^2 \times S^2$; each pair (B, W') is homeomorphic and non-diffeomorphic, distinguished by the sliceness of 4_1; and each reglued manifold is a simply connected exotic $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. (We make no claim that the smooth structures arising from different cells are distinct from one another.)

9. Verification guide

The critical steps of the derivation chain, in decreasing order of delicacy, each with the certificate to check:

(1) **The T_α framing word** $\mathrm{dir}^{\mathrm{base}}_{T_\alpha} = Ar^{-1}$ (Section 4.4). This is the Akhmedov–Park-Lemma-7-type datum. The framing identification itself is now proved (Lemma 4.1, by explicit Weinstein charts); the residual risk has narrowed to: the two Moser normalizations invoked there, the crossing-at-η conjugation formula, the identities $\varphi_0(c_y) = c_s$ and $\tilde{\varphi}(y_1) = s_1$, and the correspondence between the geometric pushoff curves of the lemma and the developed words (the engine certificates). The T^4 calibration (Section 4.6) does *not* constrain this item — trivial monodromy has no drift analogue — so it remains the least externally tested step of the chain.

(2) **The correction architecture** (Sections 4.3 and 4.5): the claim that basing the meridians along the partial-loop whiskers y_1, s_2 makes two corrections exactly $M^\pm, N^\pm$; the lasso $\delta = r^{-1}$ and its V4 identity; the whisker-sweep crossing count behind the pushoff-basing correction — one interior crossing of the $s_2 \times \bar{\beta}$ membrane with T_α, at $(P_{s'}, \mathrm{wrap})$, certified by the position table (the c-crossing precedes the e-crossing on the s-edge) and by $\bar{\beta}$’s single α-cut crossing — and its sign anti-coupling to the Bs-correction; the left-placement convention (probed by the 256-case placement sweep, but a systematic membrane-orientation error common to all placements is conceivable — such an error is now also constrained by the T^4 calibration, which it would fail). This item is exactly the error class the calibration detects, and the one place it has already found (and led to the repair of) an error.

(3) **The completeness argument** (Section 6): the two cut-surface computations (checkable by redoing the χ-arithmetic and tree contraction); the two redundancy reductions (the puncture-crossing sweep and the mapping-torus-not-product argument); the corner closure (Lemma 6.1 — check the partial-mapping-torus interface groups and the section-injectivity step). Note that completeness affects only nontrivial verdicts; the trivial verdicts need only the truth of the included relations.

(4) **The model conventions** (Section 2): the cut-square gluing directions; the identification of $\bar{\alpha}, \bar{\beta}$ holonomies with $\tilde{\varphi}, \tilde{\psi}$ (an inversion here is an ε-flip, which is swept); the octagon link walk (validated by V2, which an erroneous link structure would fail).

(5) **The engine** (Section 3): validated by V1–V4; an error here would have to survive all four validations.

The rewriting-system certificates add independent decision engines on top of the enumeration — kbmag, corroborated case-by-case by the independently implemented MAF [MAF] — with a small trusted base for the verdicts that matter here: a triviality verdict rests only on each Knuth–Bendix rule being a relator consequence (an audit of the completion trace) and on the recorded generator reductions terminating at the empty word; the confluence and language-size computations are corroborating rather than essential (`kb_certify.g`, `maf_certify.sh` and their run records, ancillary files).

An error found in any item above would itself be of interest: by the companion paper's sensitivity analysis, it would localize the exact based-loop datum on which the exotic $S^2 \times S^2$ problem turns.

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